

ALL BRANCH (Hinglish)



General Aptitude

Quantitative Aptitude

DPP 05 Discussion Notes
Mixtures Alligations



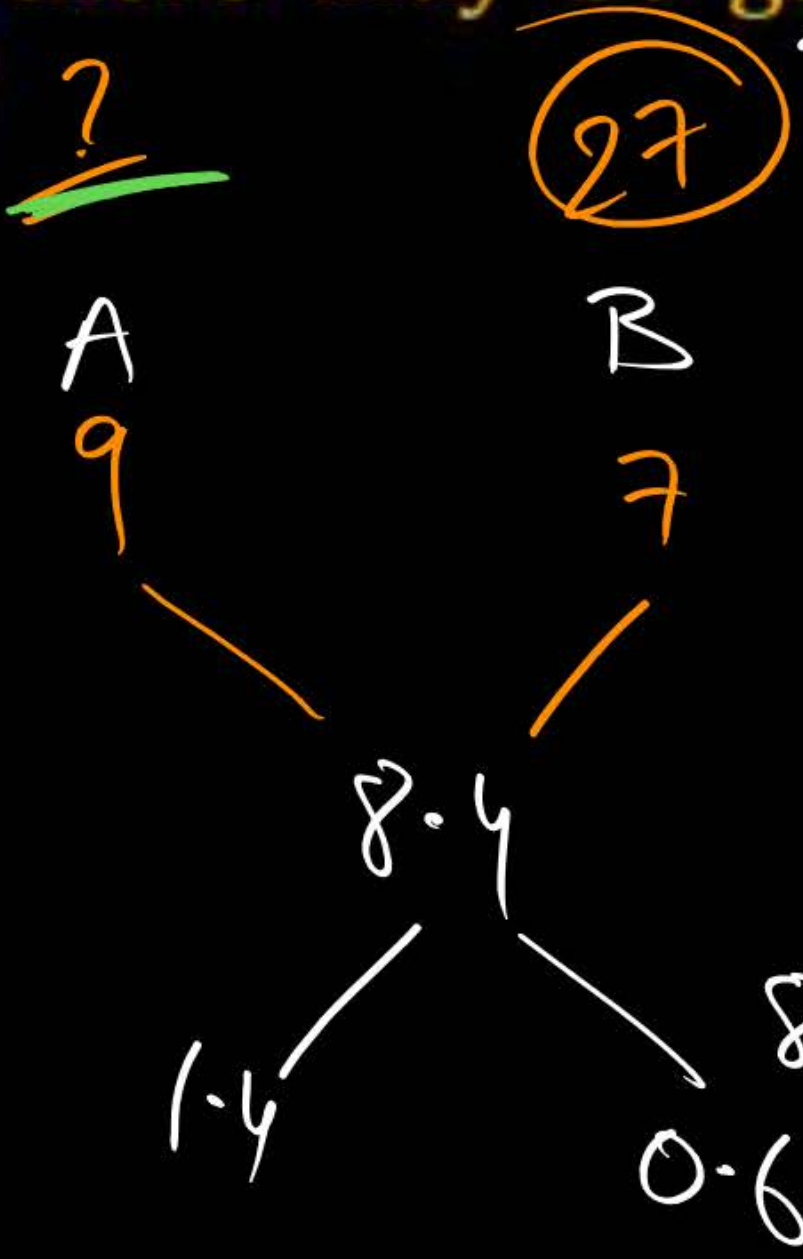
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MCQ

How many Kg of rice costing ₹9 per kg must be mixed with 27 kg of rice costing ₹7 per kg so that there may be gain of 10% by selling the mixture at ₹9.24 per kg?

- ☒ A 63 kg
- ☐ B 42 kg
- ☐ C 54 kg
- ☐ D 36 kg

$$\begin{aligned}
 A:B &= 1.4 : 0.6 \\
 &= 14 : 6 \\
 &= 7 : 3 \\
 &\quad \times 9 \quad \times 9 \\
 &= \textcircled{63} : 27
 \end{aligned}$$



$$\begin{aligned}
 S.P &= 9.24 \\
 P &= 10\% \\
 \frac{9.24}{C.P} &= 1.1 \\
 8.4 &= \frac{9.24}{1.1} = C.P
 \end{aligned}$$

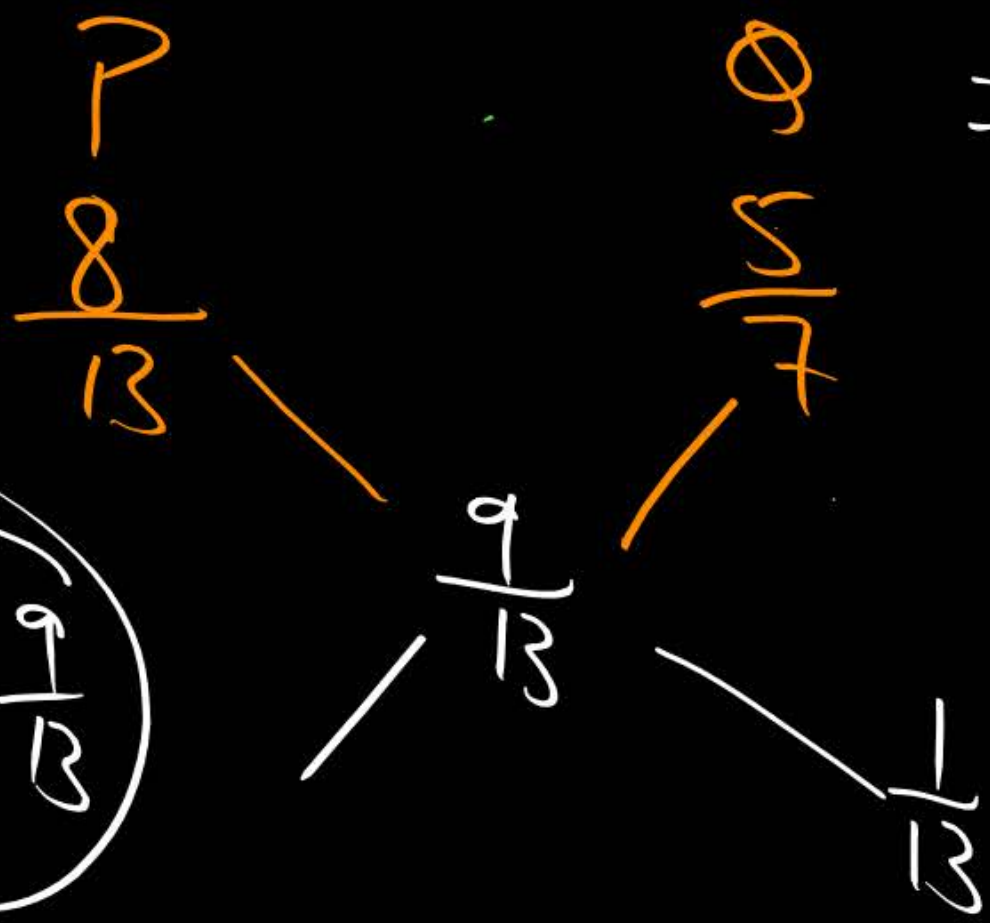
MCQ

Two vessels P and Q contain milk and water mixed in the ratio 8:5 and 5:2 respectively. In what ratio these two mixtures are to be mixed to get a new mixture containing $69\frac{3}{13}\%$ milk?

- ☒ A 2:7
- ☐ B 5:7
- ☐ C 5:2
- ☐ D 3:5

$$P:Q = \frac{2}{91} : \frac{1}{13}$$

$$= 2:7$$



$$69\frac{3}{13}\%$$

$$= \frac{900}{1300}$$

$$\frac{2}{91} = \frac{65 - 63}{91} = \frac{5}{7} - \frac{9}{13}$$

MCQ

In a mixture of 60 litres, the ratio of milk and water is 2:1. What amount of water must be added to make the ratio of milk and water as 1:2?

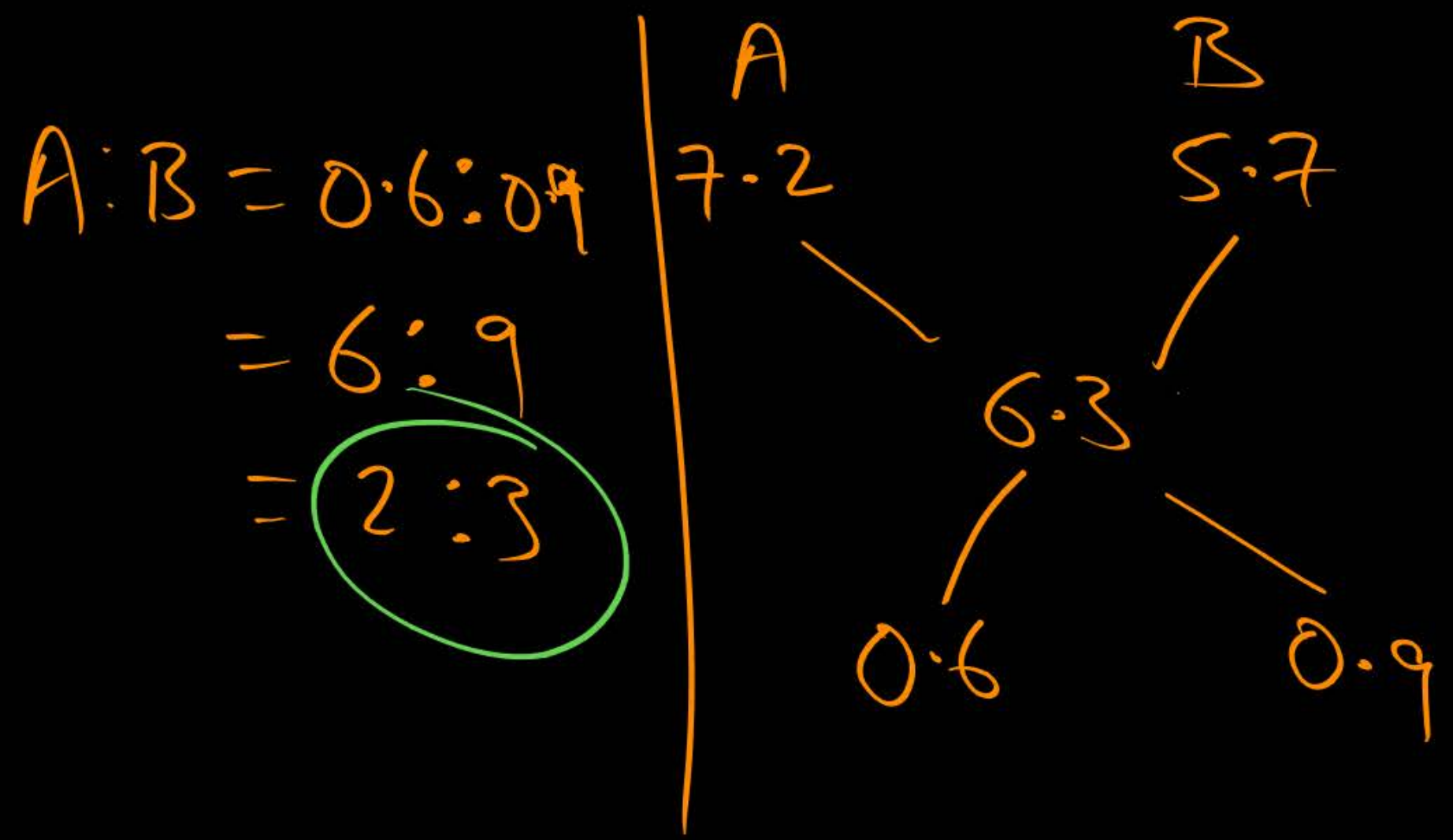
- A 56 litres
- B 42 litres
- C 60 litres**
- D 77 litres

Mix Water
 $\frac{1}{3}$
 Mix: Water
 $= \frac{1}{3} : \frac{1}{3}$
 $= 1 : 1$
 $\times 60 \quad \times 60$
 $= 60 : 60$

MCQ

Find the ratio in which sugar at ₹7.20 a kg be mixed with sugar at ₹5.70 a kg to produce a mixture worth ₹6.30 a kg.

- ☒ A 2 : 3
- ☐ B 3 : 4
- ☐ C 4 : 5
- ☐ D 1 : 3



MCQ

✓

A merchant has 1000 kg of wheat flour, part of which he sells at 8% profit and the rest at 18% profit. He gains 14% on the whole. Find the quantity he sold at 18% profit.

- A** 400 kg
- B** 560 kg
- C** 600 kg
- D** 640 kg

$$8\%P : 18\%P$$

$$= 4 : 6$$

$$= 2 : 3$$

$$\frac{3}{5} \times 1000 = 600 \text{ kg}$$



MCQ

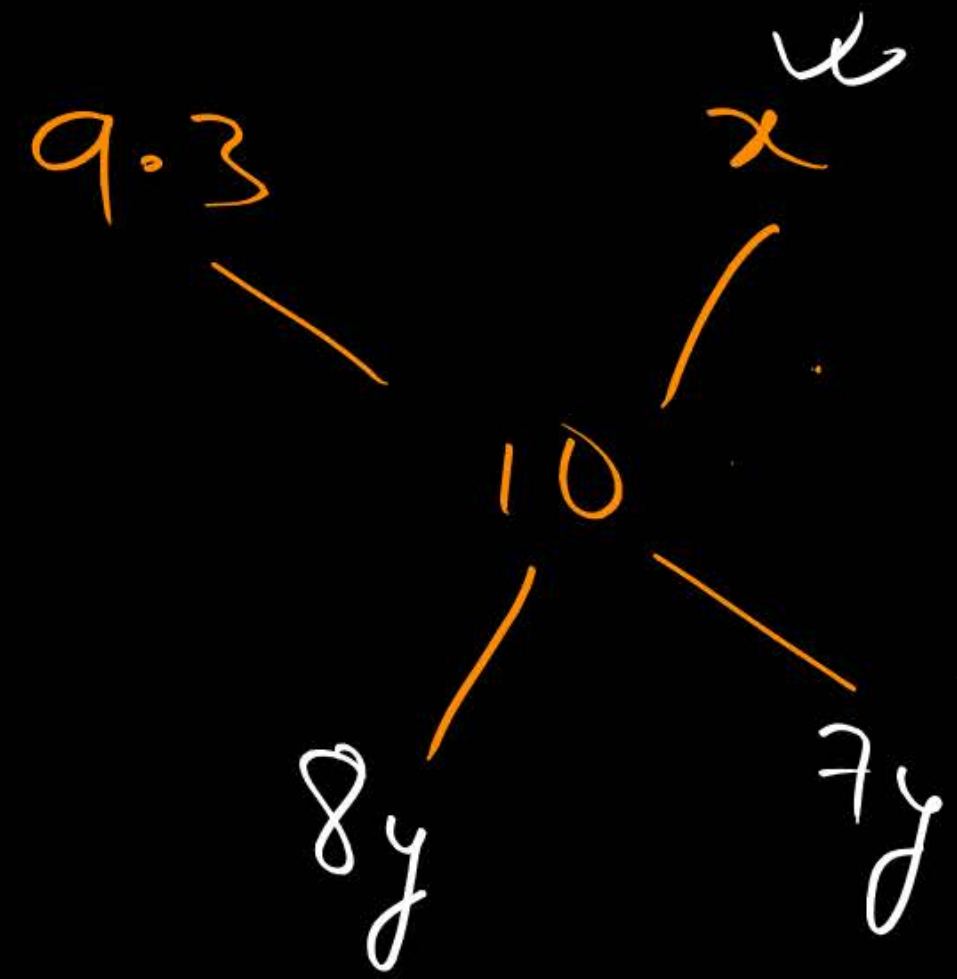
One quality of sugar at ₹9.30 per kg mixed with another quality at a certain rate in the ratio 8:7. If the mixture so formed be worth ₹10 per kg, what is the rate per kg of the second quality of sugar?

- A** ₹10.30
- B** ₹10.60
- C** ₹10.80
- D** ₹11

$$\frac{10 - 9.3}{x - 10} = \frac{7}{8}$$

$$x - 10 = 0.8$$

$$x = \underline{\underline{10.8}}$$



MCQ

A milk vendor has two cans of milk. The first contains 25% water and rest milk. The second 50% water. How much quantity should be mixed from each of the containers so as to get 12 litre of mixture where the ratio of water to milk is 3:5?

- ☐ A 4 litre; 8 litre
- ☒ B 6 litre; 6 litre
- ☐ C 5 litre; 7 litre
- ☐ D 7 litre; 5 litre

$$\begin{aligned}
 P:Q &= \frac{12.5}{25} : \frac{12.5}{50} \\
 &= 1:1 \\
 &= 6:6
 \end{aligned}$$

$$\begin{aligned}
 &\frac{3}{8} \times 100 = 37.5 \\
 &\frac{300}{8} = 37.5
 \end{aligned}$$

MCQ

Two vessels A and B contain Spirit and Water mixed in the ratio 5:2 and 7:6 respectively. Find the ratio in which these mixtures be mixed to obtain a new mixture in vessel C containing spirit and water in the ratio 8:5.

- A** 4 : 3
- B** 3 : 4
- C** 5 : 6
- D** 7 : 9

$$A:B = \frac{1}{13} : \frac{9}{91}$$

$$= 7 : 9$$

A $\frac{5}{7}$		B $\frac{7}{13}$	$\frac{65-56}{91}$ $= \frac{9}{91}$
	$\frac{8}{13}$		
$\frac{1}{13}$		$\left(\frac{5}{7} - \frac{8}{13}\right)$	
		$\frac{9}{91}$	

MCQ

In a mixture of 48 litre, the ratio of milk and water is 2 : 1. if this ratio is to be 1 : 3 then what is the quantity of water to be further added?

- ☐ A 96 litre
- ☐ B 72 litre
- ☐ C 60 litre
- ☒ D 80 litre

Milk: water =

$$\frac{1}{4} : \frac{5}{12}$$

$$= 3 : 5$$

$\times 16$ $\times 16$
48 : 80

Milk Water

$$\frac{1}{3} \quad \quad \quad 1$$

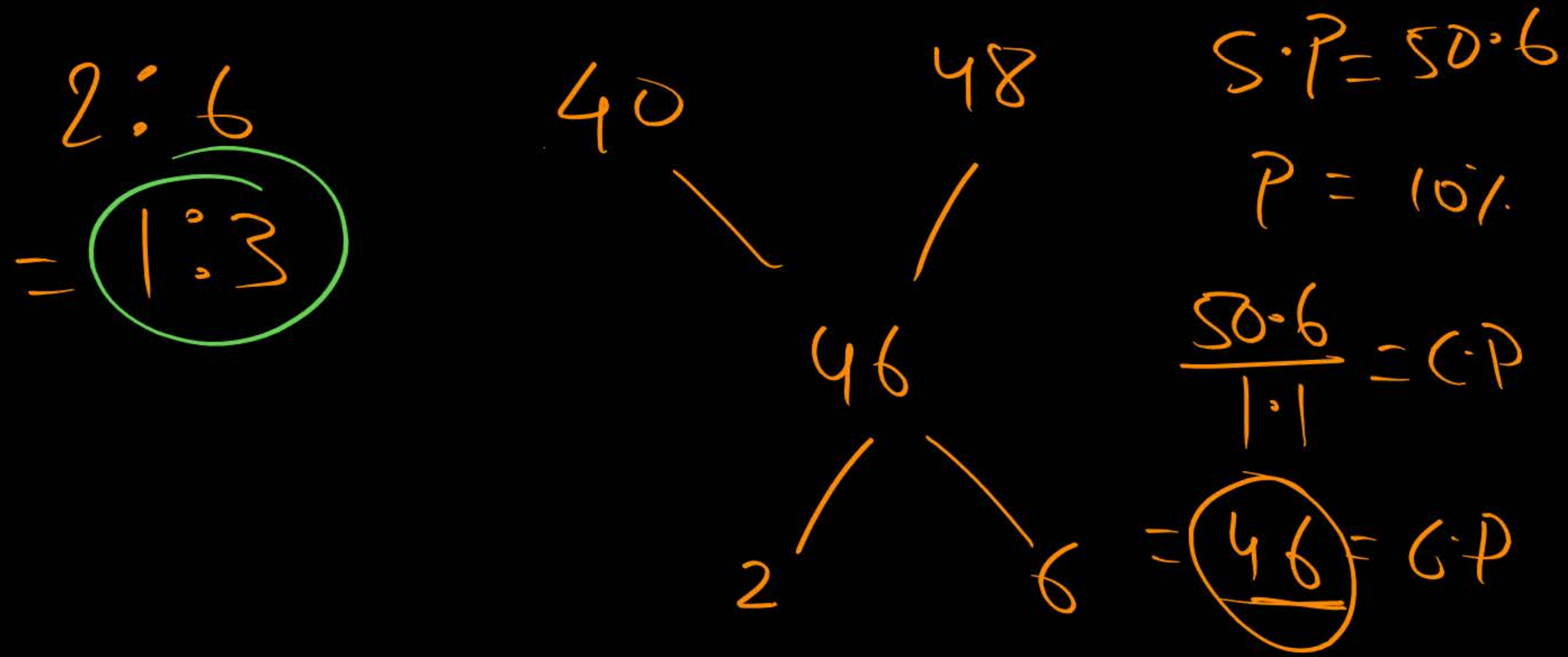
$$\frac{1}{4} \quad \quad \quad \frac{9-4}{12} = \frac{5}{12}$$

$$\frac{1}{4} \quad \quad \quad \left(\frac{3}{4} - \frac{1}{3} \right) = \frac{5}{12}$$

MCQ

In what ratio must a grocer mix two varieties of rice worth ₹40 per kg and ₹48 per kg so that by selling the mixture at ₹50.6 per kg he may gain 10%?

- ☐ A 1 : 2
- ☒ B 1 : 3
- ☐ C 2 : 3
- ☐ D 4 : 1





Thank You!

GW Soldiers